CHE 2201 – Physical Chemistry II
Phase Equilibria

Lec. 2. One Component Systems

2.1. Phase Diagram of a One Component System

In one component systems, the separate phases are the same substance in different aggregate states. If a substance can have different crystalline modifications, each of them forms a separate phase. Thus, water forms six different modifications of ice, sulphur crystallizes in the rhombic and monoclinic forms etc. Each of the above modifications is stable within definite intervals of T and p.

The maximum number of phases possible in a one component equilibrium system can be found using the Phase Rule:

$$F = C + 2 - P \tag{2.1}$$

where F – number of degrees of freedom;

C – number of components;

P – number of phases.

However, the number of degrees of freedom (F) could be equal to zero or to a + integer, i.e. $F \geq 0$ (0, 1, 2, 3, …..).

It follows that:

$$C + 2 - P \geq 0 \tag{2.2}$$

$$P \leq C + 2 \tag{2.3}$$

For a one component system, C = 1, and so, we have

$$P \leq 3 \tag{2.4}$$

(i.e. P = 1, 2 or 3).

Accordingly, no single substance can form an equilibrium system consisting of more than three phases. This conclusion is completely confirmed by experiment.

In the general case, we do not know the form of the equations of state of different phases of both multi-component and one-component systems. The only exception is the Clapeyron-Mendeleev equation, which is applicable when the components of gaseous phase obey the laws of ideal gas. There are a number of complicated equations describing the state of real gases and individual liquids.

The possibility of finding the relationship between the values of variables determining the state of a system, is the method of direct measurements of T, p and concentrations (or volumes) of components. The data obtained by these measurements are used to construct the phase diagrams, which graphically express the laws being sought.

So, the phase diagram of a substance is a map showing the conditions of temperature and pressure at which its various phases are thermodynamically stable.

The equation of state for a one component system include three variables, for example, temperature (T), pressure (p) and the concentration – C (or molar volume – V). Any two of them can be considered as independent variables, and the third one as their function. Usually, T and p are taken as independent variables.

By laying off the values of these two variables along the two axes, we obtain a two-dimensional phase diagram, as shown in Fig. 2.1. Each point on this diagram expresses the conditions, which the system is in. This makes it possible to divide the entire two-dimensional diagram into several areas, each of which includes all the possible combinations of T and p corresponding to the equilibrium existence of a definite phase.

Thus, in the figure 2.1.a, area G corresponds to the conditions of existence of the gaseous phase, L – liquid phase and S – solid phase. The points reflecting the state and conditions of existence of a system are called figurative points.

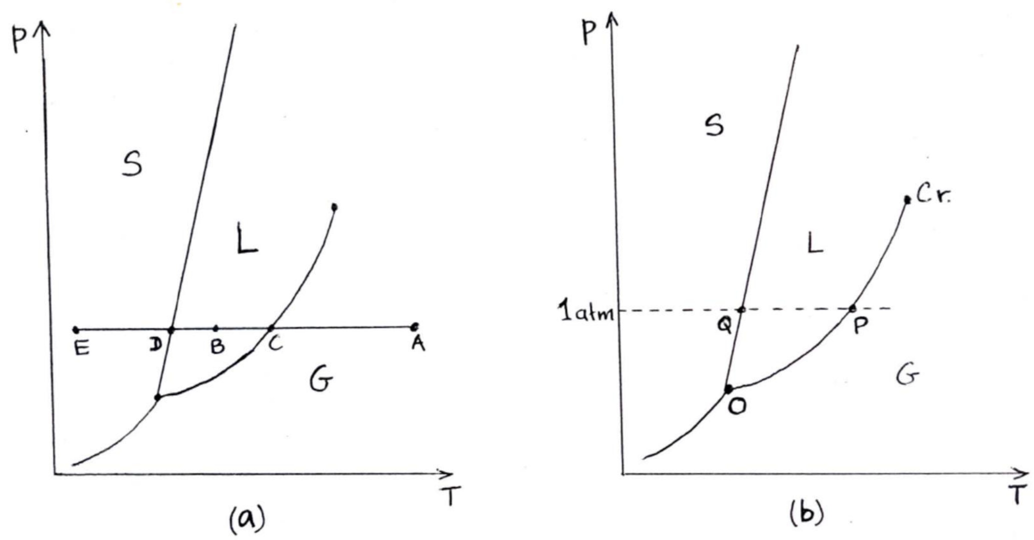

Fig. 2.1. (a). Typical 2-Dimensional Phase Diagram of a One Component System; (b). Significant Points of a Phase Diagram

Figure 2.1.a:

At point A, the gaseous phase of the substance is the thermodynamically most stable phase. At point B, the liquid phase is the most stable one. If the system is arranged to have a p and T represented by point C, then the liquid and its vapour are in equilibrium. If the T is reduced at constant p, the system moves to the point B, where only liquid is present. If the T is reduced still further to D, then the solid and the liquid phases can coexist in equilibrium. A further reduction in T to the point E, takes the system into the region where the solid alone is present.

The boundaries between regions in a phase diagram, which are called phase boundaries, show the values of p and T at which the two neighbouring phases are in equilibrium. Any point lying on a phase boundary represents p and T at which there is a dynamic equilibrium between the two neighbouring phases. A state of dynamic equilibrium is one in which a reverse process is taking place at the same rate as the forward process. Although there may be a great deal of activity at a molecular level, there is no net change in the bulk properties or appearance of the sample.

Figure 2.1.b:

The T at which the vapour pressure of a liquid is equal to the external pressure is called the boiling temperature. When the external p is1 atm. the boiling temperature is called the normal boiling point. Thus, the point P in Fig. 2.1.b represents the normal boiling point.

The T at which the liquid and solid phases of a substance coexist in equilibrium is called the melting temperature of the substance. Because a substance melts at the same T as it freezes, the melting T is the same as the freezing temperature. The melting T, when the external p is equal to 1 atm, is called the normal melting T. Thus, the point Q in Fig. 2.1.b represents the normal melting point or normal freezing point.

There is a set of conditions, under which the three different phases (typically solid, liquid and vapour), all simultaneously coexist in equilibrium. It is represented by the triple point (point O in the Fig. 2.1.b), where the three phase boundaries meet. The triple point of a pure substance is its characteristic physical property. For example, for water, the triple point lies at 273.16 K and 611 Pa. At the triple point, there is a dynamic equilibrium between all three phases.

Now let us consider what happens when we heat the liquid in a closed vessel. Under such conditions, the liquid does not boil. Instead, the density of vapour increases as the vapour pressure increases, because the vapour cannot escape (it is confined to a fixed volume). In due course (at a certain T), the density of the vapour becomes equal to that of the remaining liquid. At this stage the surface between the two phases disappears. The T at which the surface disappears is the critical temperature (T_C). The vapour pressure at the critical T is called the critical pressure (p_C), and the critical T and critical p together identify the critical point of the substance (point Cr in the Fig. 2.1.b). At and above T_C, a single uniform phase (gaseous) fills the container. Above T_C, liquid does not exist. That is why the liquid-vapour phase boundary terminates at critical point.

Thus, the triple point marks the lowest T at which the liquid can exist, while the critical point represents the highest T at which the liquid can exist.

2.2. Phase Diagrams of Water and Sulphur

The two-dimensional phase diagram for water is shown schematically in Fig. 2.2.

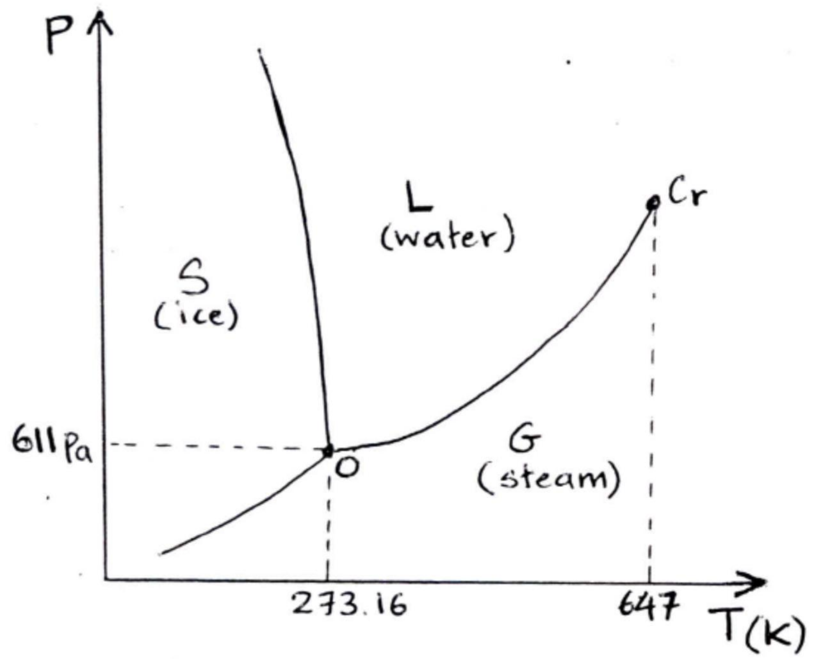

Fig. 2.2. Phase Diagram of Water (Not drawn to the scale)

The triple point of the equilibrium coexistence of vapour (steam), water and ice corresponds to the pressure of 611 Pa and 273.16 K. It has to be noted that six different modifications of ice are not considered in this phase diagram.

Crystalline sulphur can exist in two different modifications, rhombic and monoclinic. Therefore, sulphur forms four phases, two crystalline ones, a liquid and a vapour. The phase diagram of sulphur is shown schematically in Fig. 2.3.

The phase boundaries divide the diagram into four areas corresponding to the conditions of the equilibrium states of vapour, liquid and two crystalline modifications. The lines themselves correspond to the conditions in which the equilibrium coexistence of two phases is possible.

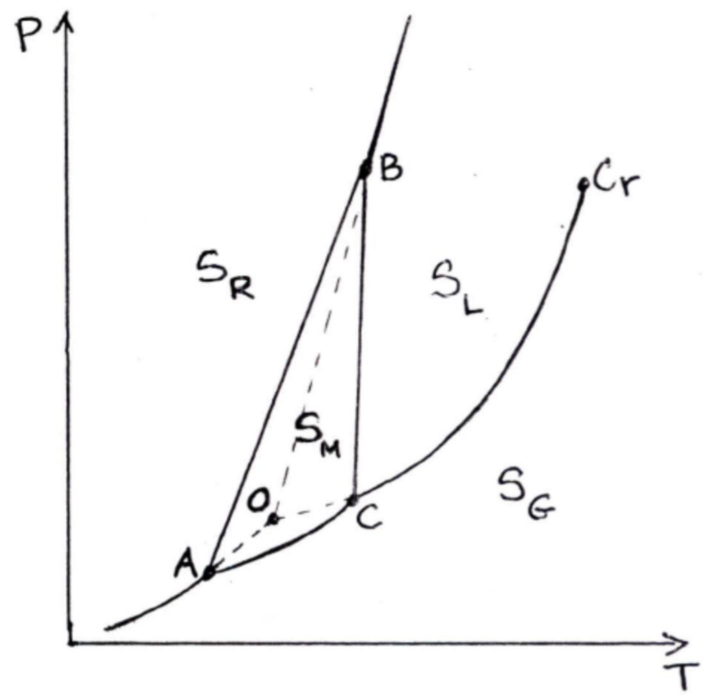

Fig. 2.3. Phase Diagram of Sulphur (Not drawn to the scale)

At points A, B and C, three phases are in thermodynamic equilibrium. In addition, there is another triple point O, at which superheated rhombic sulphur, supercooled liquid sulphur and vapour supersaturated relative to monoclinic sulphur, can coexist. The chemical potentials of the three phases at the T and p corresponding to the point O are the same. As a result, the three thermodynamically non-equilibrium phases can form a metastable system, i.e. a system that is in a state of relative stability.

2.3. Three-Dimensional Phase Diagrams

As mentioned earlier, the equations of state for a one component system include three variables, i.e. T, p and C or T, p and V. Any two of them can be considered as independent variables.

To show graphically the relations between the values of T, p and C (or V), we must use a system of coordinates including three mutually perpendicular axes, each corresponding to the values of one variable. In other words, we have to construct 3-D phase diagrams. Figure 2.4 shows as an example, a 3-D phase diagram of CO_2 without observance of the proper scale.

The complete 3-D phase diagrams are not in favour in day-to-day work, as they are inconvenient in use, and their preparation involves lot of work. The projections of a complete diagram on the planes passing through the coordinate axes do not have these shortcomings.

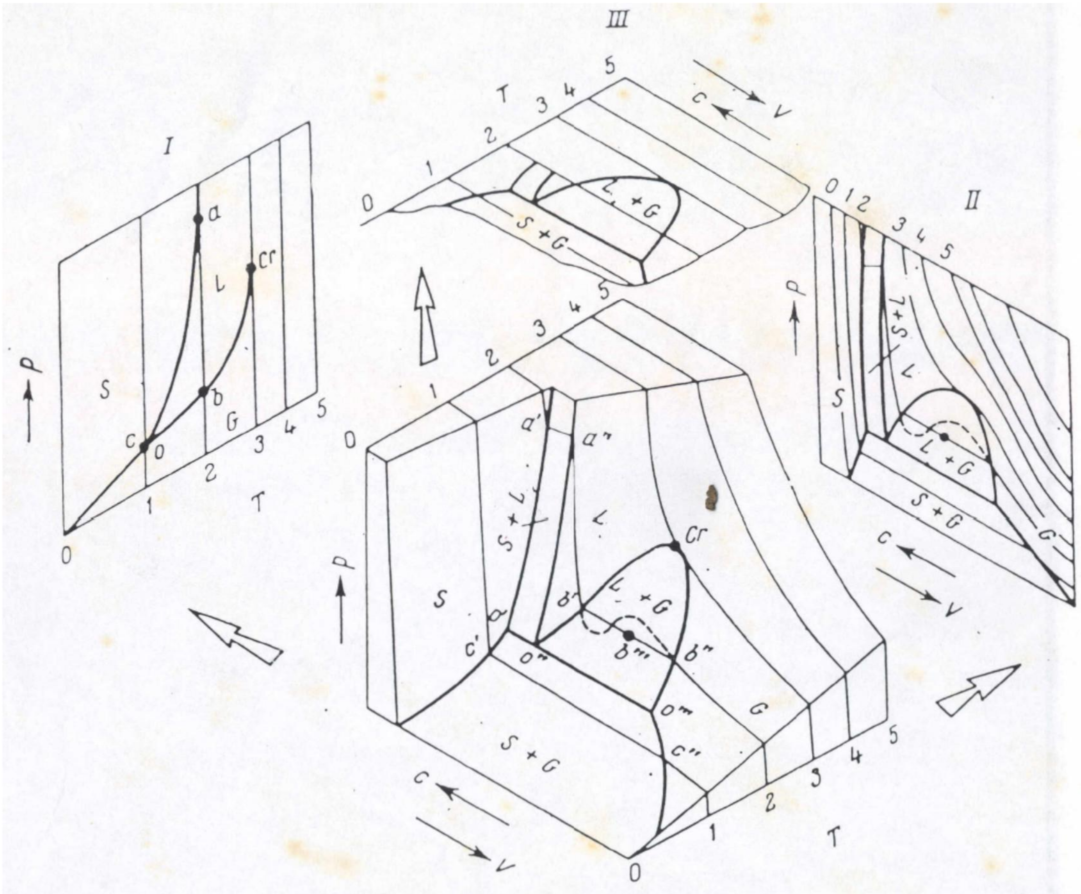

Fig. 2.4. 3-D Phase Diagram of CO_2 and Its Projections on the Planes p-T, p-V & V-T